STDISCM Problem Set 1
Multithreaded Matrix Multiplication [50 pts]

Background

Let A be an $m\ x\ p$ matrix and B be a $p\ x\ n$ matrix. Their product $C = AB$ is an $m\ x\ n$ matrix where each element at the i th row and j th column can be computed as

$$c_{ij} = \sum_{k=1}^{p} a_{ik}\, b_{kj}$$

for $i = 1,\ 2,\ \ldots,\ m$ and $j = 1,\ 2,\ \ldots,\ n$. Essentially, the resulting matrix is composed of the dot products of each row in A and each column in B.

The formula above has a time complexity of $O(mpn)$, or $O(n^3)$ if both A and B are square matrices. Thus, matrix multiplication is computationally expensive, especially for large matrices.

Objective

Your goal is to write a C++ program that uses concurrency to reduce the time needed to perform matrix multiplication. External libraries (BLAS, LAPACK) cannot be used for this task, so you must write your own functions for computing the product of matrices.

Your program should accept an input file (`input.txt`) that contains two matrices, separated by a line break. The elements within each matrix are separated by a single space, with each row of text corresponding to a row in the matrix. The program will then output a file (`output.txt`) containing the product of the given matrices. Both the input and output file values must be limited to a single decimal place. The maximum dimension that your program should allow is $1000\ x\ 1000$.

Additionally, your program must include a standard implementation of matrix multiplication. You should measure and display the computation time for both the standard and multithreaded versions in microseconds. The result of the standard implementation does not need to be saved to the output file.

Sample Input File

```
1.2 3.5
5.6 7.3
8.2 1.9

2.5 1.7 3.8
5.4 6.1 0.9
```

Sample Output File

```
21.9 23.4 7.7
53.4 54.0 27.8
30.8 25.5 32.9
```

Sample Console Output

```
Standard Running time: 7 µs
Multithreaded Running time: 248 µs
```

Report

Write a one-page PDF report that explains your multithreaded approach and shows benchmarking results that compare the running times of both solutions for various matrix sizes. You should also indicate at which matrix size the multithreaded solution starts to gain performance advantage and explain why the standard formula might run faster for certain inputs.

Submission

Via a Canvas online quiz